

Tab 1

Bluestock Mutual Fund Analytics

Capstone Project Report

Prepared by: Shivani Sharma

Internship: Bluestock Data Analytics Internship

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Executive Summary

This project analyzes Indian mutual fund industry data to understand fund performance, investor behavior, risk metrics, and market trends.

The project covers:

- Data cleaning and ETL pipeline creation
- SQLite based analytical database
- Exploratory Data Analysis
- Fund performance evaluation
- Risk analysis
- Investor analytics
- Power BI dashboard development

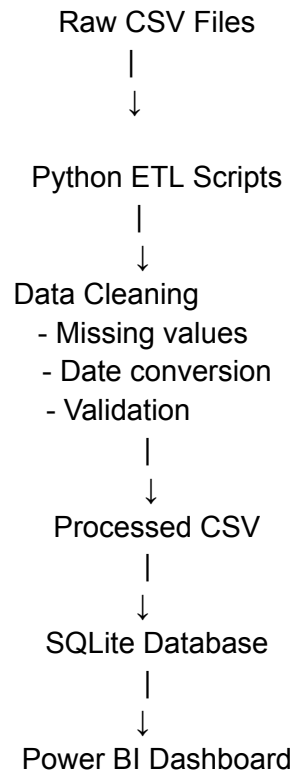
The final solution provides actionable insights for investors, fund managers, and analysts.

Data Sources

Dataset	Description
nav_history.csv	Daily NAV information
investor_transactions.csv	Investor SIP/Lumpsum transactions
scheme_performance.csv	Fund returns and expenses
Portfolio_holdings.csv	Sector allocation
nifty100.csv	Benchmark index data

Source: *Bluestock Mutual Fund Dataset*

ETL Design



ETL Pipeline:

Extract:

Loaded raw CSV datasets using pandas.

Transform:

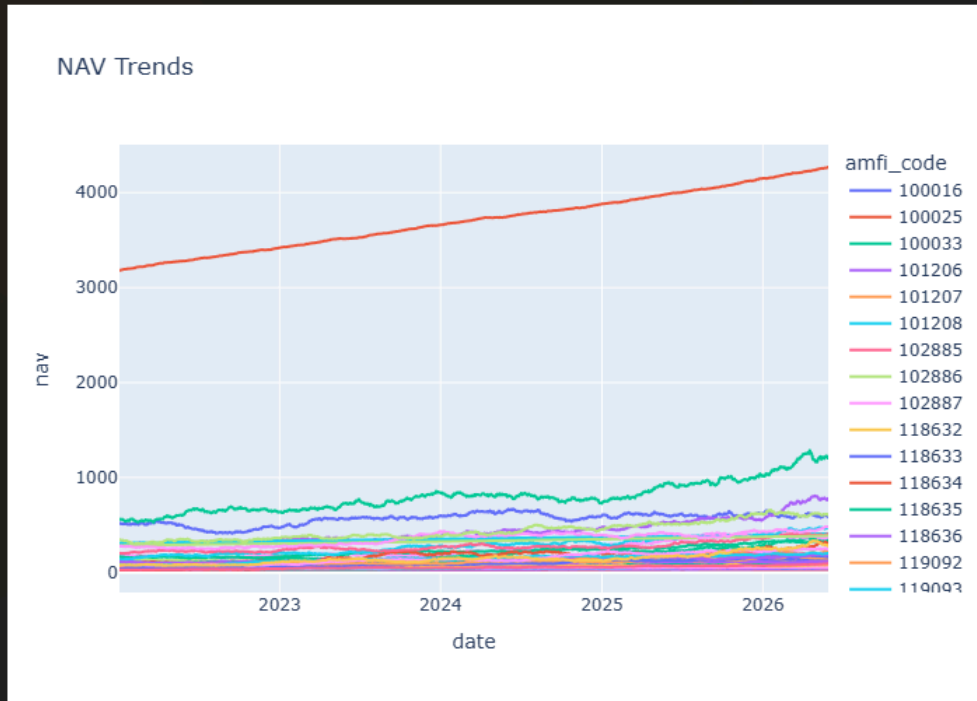
Applied validation, formatting and calculations.

Load:

Stored cleaned data into SQLite database.

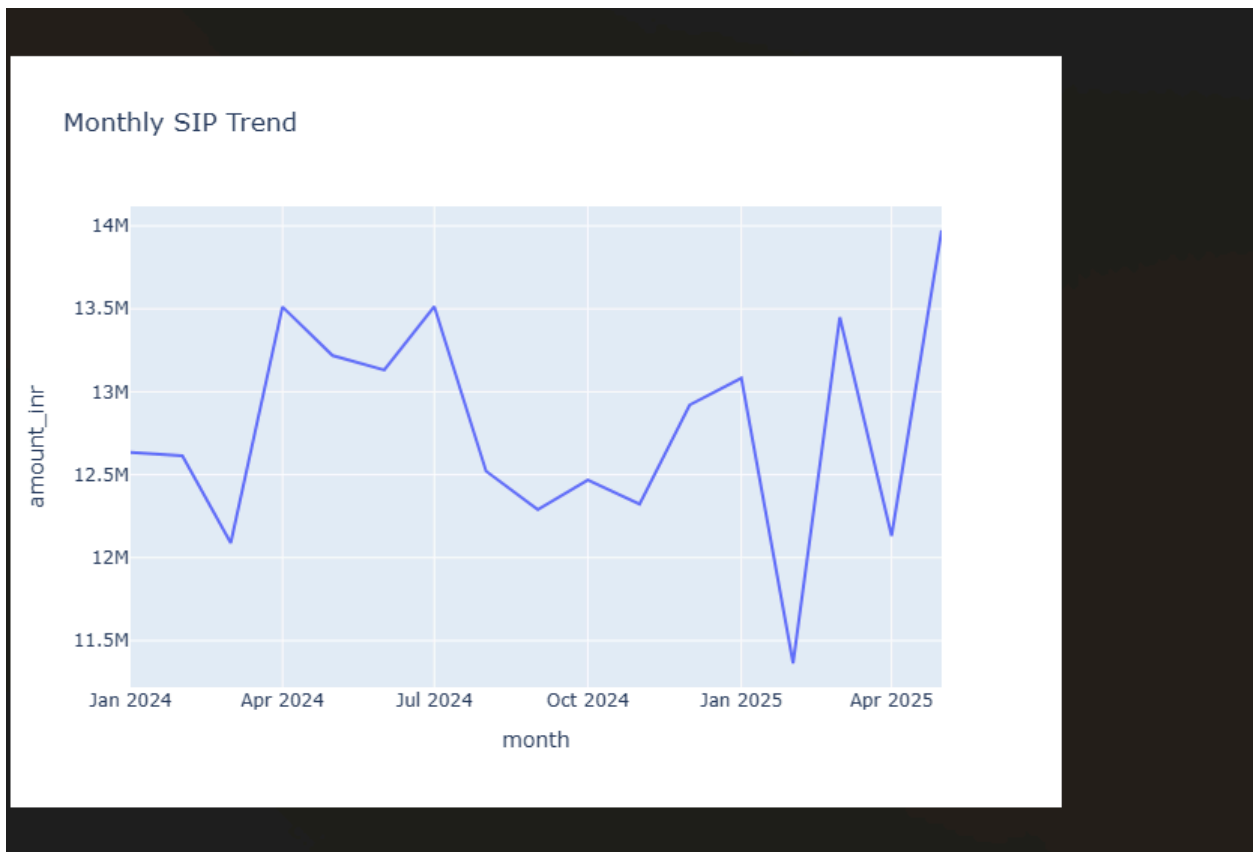
EDA Findings

NAV Trends:



Insights: NAV values showed consistent long-term growth with short-term market corrections.

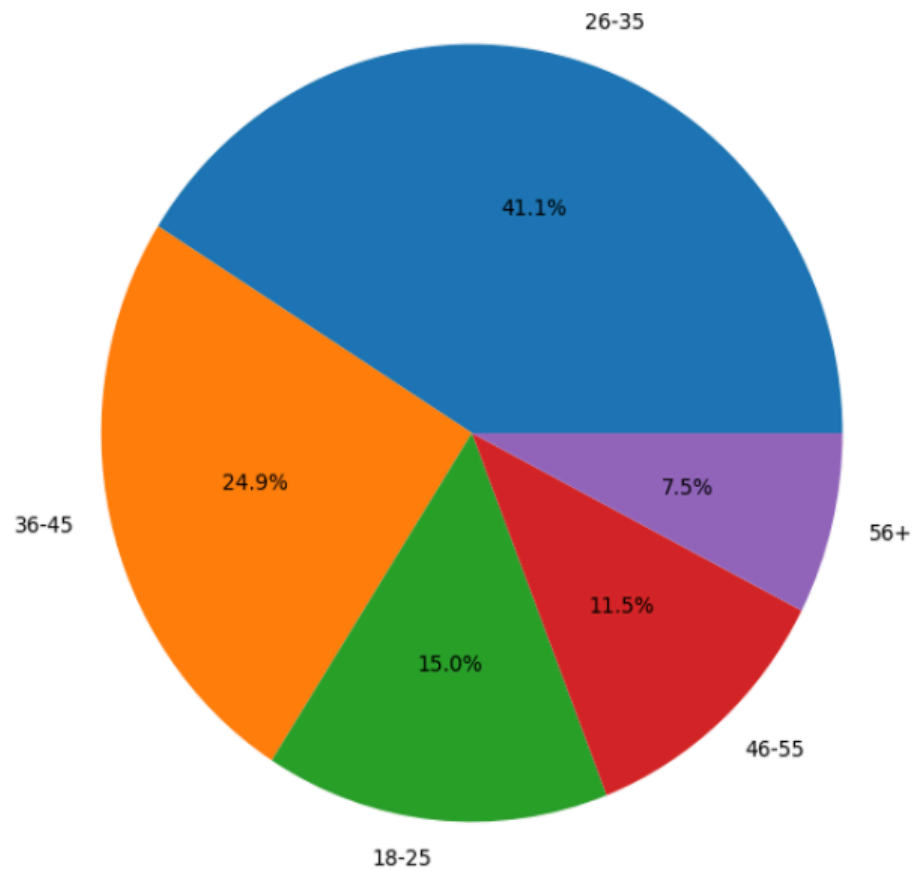
SIP Trends:



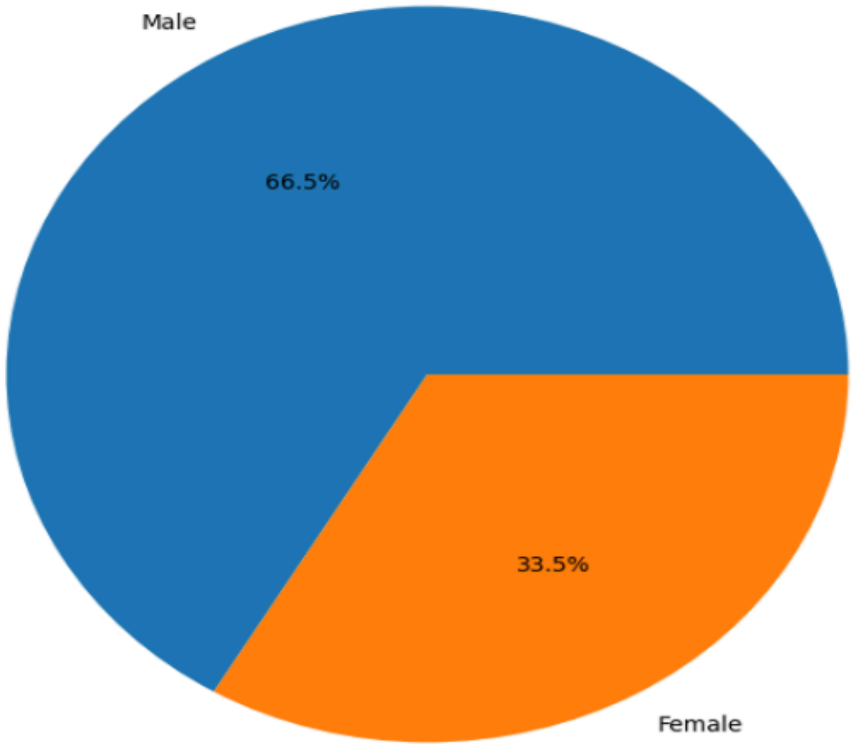
Insights : SIP participation increased steadily indicating growing retail investor adoption

Investor Demographics :

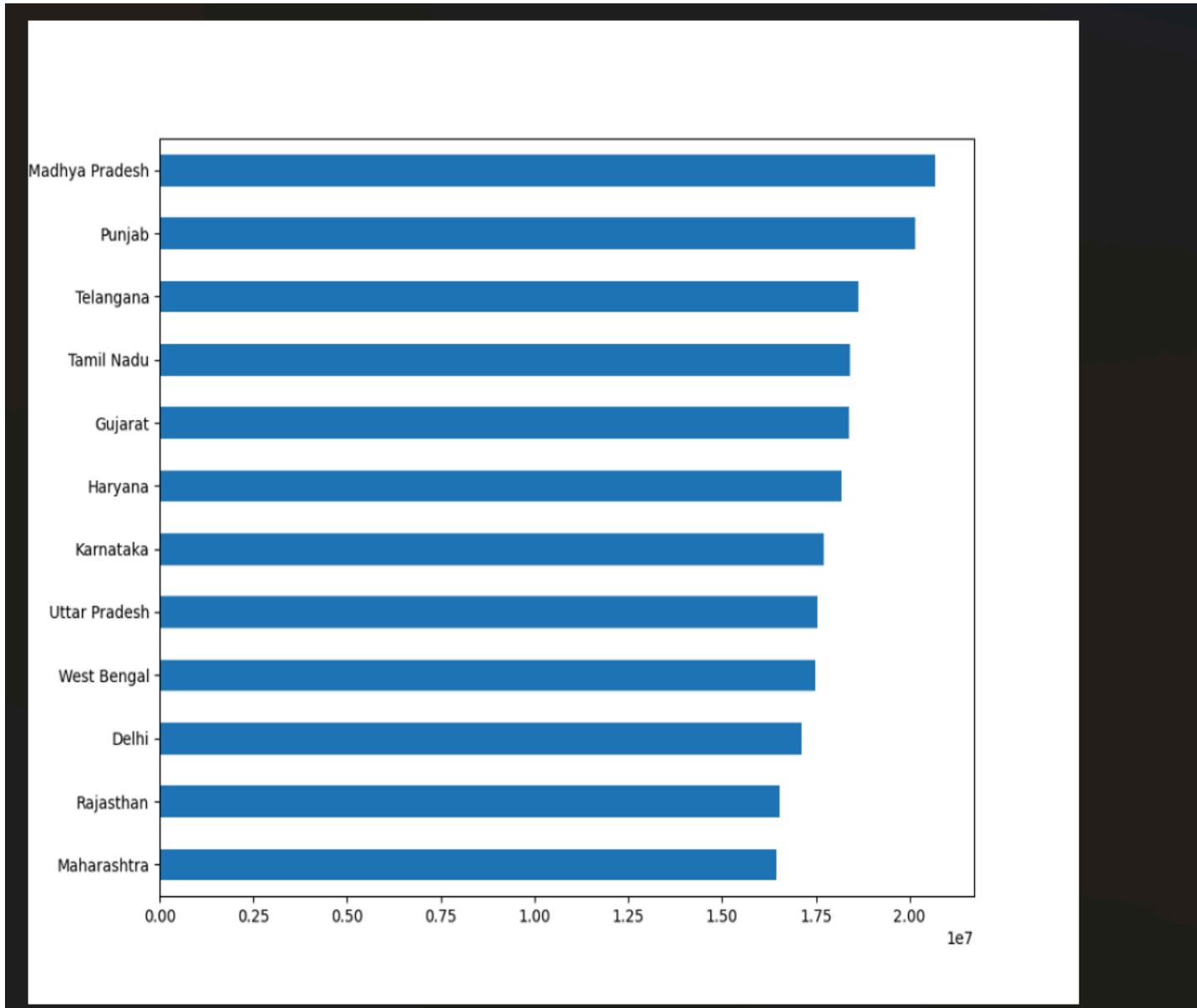
Age Distribution:



Gender Split:



State Contribution:



Performance Analysis

amfi_code	cagr	sharpe	alpha	beta	max_drawdown	return_rank	sharpe_rank	alpha_rank	dd_rank	score
100016	0.026352	-0.20152	0.059619	0.056455	-0.247344409	0.1	0.15	0.175	0.85	18.75
100025	0.044551	-0.56709	0.054792	-0.01094	-0.043082638	0.125	0.05	0.1	0.1	8
100033	0.300997	1.093699	0.233664	-0.00259	-0.162172086	0.85	0.85	0.8	0.5	67.75
101206	0.235205	1.027213	0.141329	0.034864	-0.112915963	0.725	0.8	0.475	0.225	53.5
101207	0.079331	0.162661	0.129441	0.110101	-0.354469165	0.35	0.325	0.4	0.95	36.125
101208	0.065044	-0.81557	0.06046	-0.00122	-0.001622498	0.225	0.025	0.2	0.075	12.125
102885	0.182188	0.817099	0.15132	0.025324	-0.108598622	0.575	0.625	0.55	0.175	45.625
102886	0.011709	-0.20583	0.042587	0.014874	-0.280011236	0.025	0.125	0.075	0.875	14.125
102887	0.168287	0.619518	0.152353	0.026459	-0.215398271	0.525	0.475	0.6	0.75	47.125
118632	0.240312	1.081659	0.204652	0.009527	-0.174140751	0.75	0.825	0.725	0.575	63.375
118633	0.162622	0.645207	0.151597	-0.00609	-0.186297431	0.475	0.5	0.575	0.65	44.75
118634	0.163614	0.448434	0.155673	-0.03719	-0.233448861	0.5	0.375	0.625	0.8	44.875
118635	0.159455	0.664857	0.131021	0.007338	-0.116506245	0.45	0.55	0.425	0.25	38.25

11863 6	0.053 132	-0.35 663	0.057 314	-0.00 858	-0.083163 618	0.15	0.075	0.15	0.15	10. 875
11909 2	0.063 635	0.030 785	0.068 362	-0.04 514	-0.144016 179	0.2	0.25	0.3	0.37 5	22
11909 3	0.078 919	0.129 614	0.076 518	0.008 018	-0.217513 964	0.325	0.3	0.325	0.77 5	31. 5
11909 4	0.281 926	0.998 231	0.241 069	-0.03 459	-0.209608 839	0.8	0.75	0.825	0.7	66. 25
11909 5	0.015 205	-0.07 597	0.107 647	0.021 002	-0.516777 537	0.05	0.2	0.375	0.97 5	23. 75
11912 0	0.058 853	-0.22 658	0.067 335	0.004 401	-0.043287 289	0.175	0.1	0.25	0.12 5	14
11955 1	0.257 849	1.208 267	0.196 791	0.052 42	-0.150123 855	0.775	0.925	0.7	0.42 5	64. 625
11955 2	0.215 08	0.953 279	0.186 736	0.022 817	-0.118034 935	0.65	0.725	0.65	0.27 5	53. 375
11959 8	0.323 981	0.945 308	0.290 381	-0.06 641	-0.287060 057	0.975	0.675	0.95	0.9	74. 125
11959 9	0.020 531	-0.05 719	0.001 106	0.052 399	-0.525742 212	0.075	0.225	0.025	1	18. 375
12050 3	0.187 713	0.797 973	0.189 47	0.023 891	-0.138978 628	0.6	0.6	0.675	0.35	50
12050 4	0.232 774	1.026 524	0.251 109	-0.00 767	-0.125882 756	0.7	0.775	0.85	0.3	60. 375
12050 5	0.328 016	1.180 101	0.285 787	0.018 188	-0.181885 136	1	0.9	0.925	0.62 5	77. 25
12050 6	0.170 722	0.648 879	0.144 451	0.020 627	-0.198256 583	0.55	0.525	0.5	0.67 5	46. 375
12050 7	0.072 324	0.495 723	0.067 355	-0.00 188	-0.000977 314	0.3	0.45	0.275	0.02 5	26
12084 1	0.135 223	0.494 273	0.056 887	0.029 558	-0.175736 142	0.425	0.425	0.125	0.6	31. 875

12084 2	0.066 968	0.076	0.007 787	-0.06 841	-0.240035 113	0.25	0.275	0.05	0.82 5	23. 625
12084 3	0.308 833	1.306 744	0.259 556	-0.05 015	-0.129739 682	0.875	0.975	0.875	0.32 5	71. 375
12084 4	0.069 104	-0.08 875	0.064 422	0.000 74	-0.001162 933	0.275	0.175	0.225	0.05	17. 625
12549 7	0.190 036	0.775 79	0.144 555	0.031 05	-0.152000 131	0.625	0.575	0.525	0.45	48. 125
12549 8	0.112 937	0.300 612	0.085 113	0.018 235	-0.211172 796	0.375	0.35	0.35	0.72 5	34. 25
14856 7	0.309 499	1.448 291	0.292 382	0.016 402	-0.112657 291	0.9	1	0.975	0.2	73. 5
14856 8	0.224 998	0.926 287	0.226 213	-0.00 467	-0.152725 942	0.675	0.65	0.75	0.47 5	56. 25
14856 9	0.319 245	1.234 93	0.226 913	0.022 806	-0.163967 429	0.925	0.95	0.775	0.52 5	72. 25
14932 2	0.133 835	0.464 037	0.134 798	0.010 299	-0.148445 769	0.4	0.4	0.45	0.4	35
14932 3	0.295 581	1.132 122	0.270 396	0.000 524	-0.172481 058	0.825	0.875	0.9	0.55	70. 125
14932 4	0.322 621	0.949 796	0.303 951	-0.04 042	-0.311719 003	0.95	0.7	1	0.92 5	75. 25

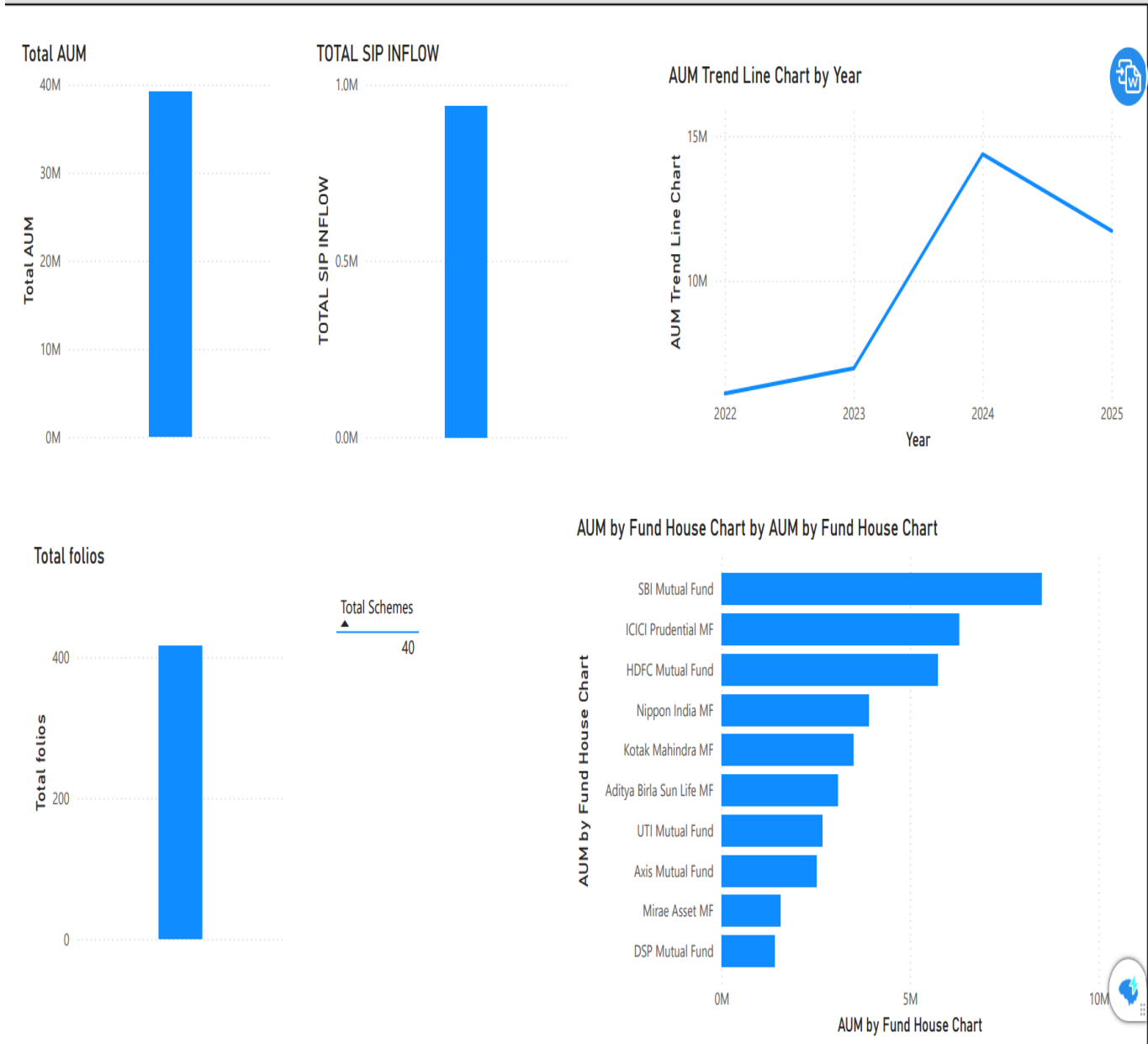
- VaR
- CVaR
- Maximum Drawdown
- Alpha
- Beta
- Sharpe ratio

Sharpe Ratio:

Sharpe ratio measures return generated per unit of risk
Higher Sharpe indicates better risk-adjusted performance.

Dashboard Screenshots

Industry Overview:



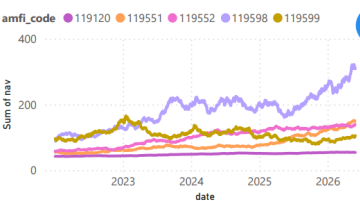
Fund Performance:

Return, Risk and Return by amfi_code



amfi_code	Sum of alpha	Sum of beta	Sum of benchmark_3yr_pct	Sum of
119599	1.13	1.04	22.01	
119598	1.23	0.89	22.16	
119552	1.78	0.87	9.52	
119551	0.87	0.89	11.49	
119120	1.60	0.22	4.47	
Total	6.61	3.91	69.65	

Sum of nav by date and amfi_code



fund_house

- ☐ Aditya Birla Sun Life MF
- ☐ Axis Mutual Fund
- ☐ DSP Mutual Fund
- ☐ HDFC Mutual Fund
- ☐ ICICI Prudential MF
- ☐ Kotak Mahindra MF
- ☐ Mirae Asset MF
- ☐ Nippon India MF
- ☒ SBI Mutual Fund
- ☐ UTI Mutual Fund

category

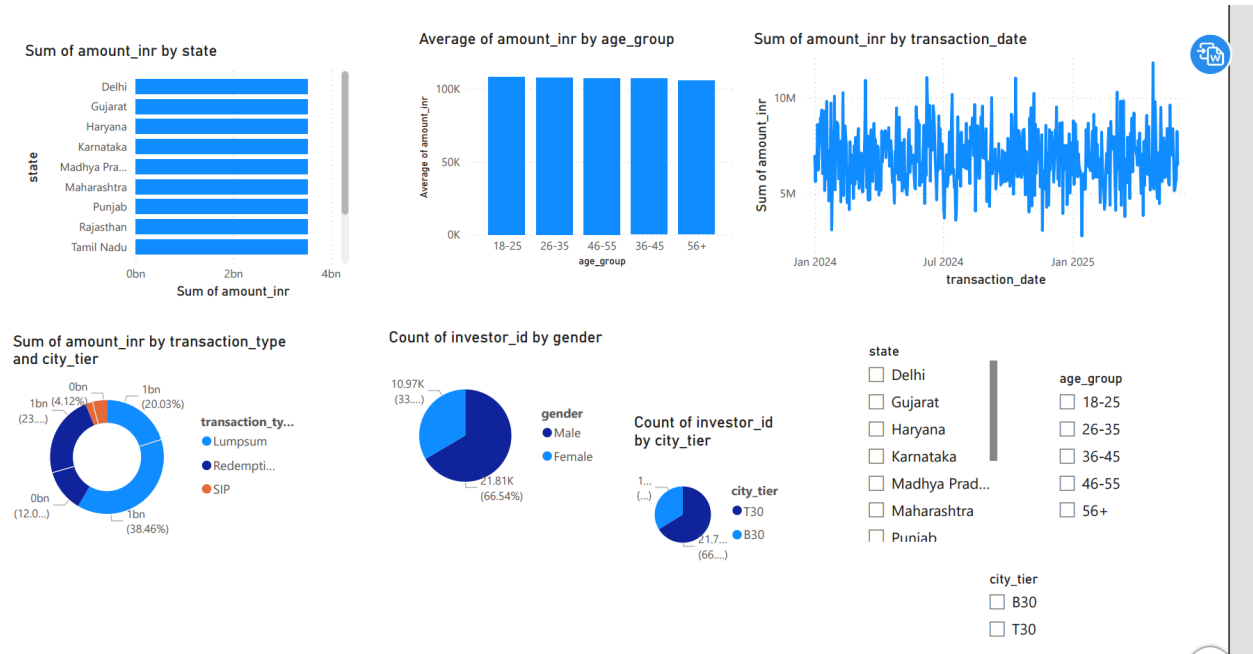
- ☐ Debt
- ☐ Equity

plan

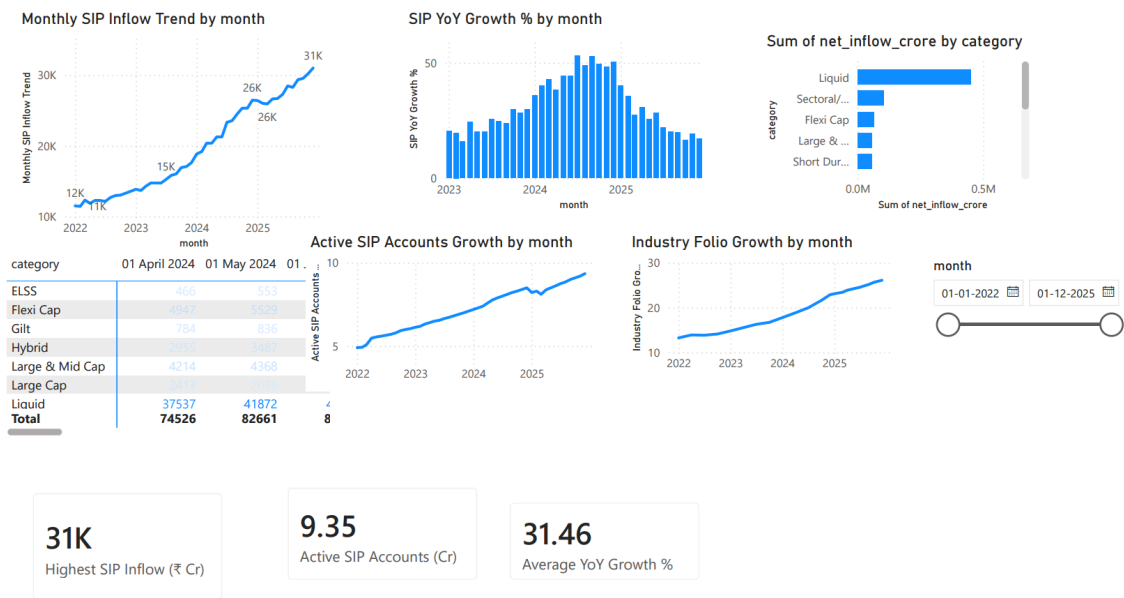
- ☐ Direct
- ☐ Regular



Investor Analytics:



Market Trends:



Limitations

1. Historical data availability was limited.
2. Benchmark comparison used Nifty 100 proxy.
3. Future market conditions cannot be predicted.
4. Investor demographic data may contain sampling limitations.

Recommendations

1. Investors should evaluate Sharpe ratio along with returns.
2. Avoid highly concentrated portfolios.
3. Maintain SIP discipline for long-term wealth creation.
4. Compare funds against suitable benchmarks.

